BLOOM: BMI & Female Health Tracker

Technical Document

A desktop application written in Python with CustomTkinter

COMPE 02: Programming Logic and Design

Members
Justin Blaire M. Ojos
Vena Lee B. Rebaula
Lesther Lazarte
Hayden Tamayo

TABLE OF CONTENTS

1. Introduction
2. Objectives, Scope and Limitations
3. Requirements and System Specifications
4. Program Structure and Code Organization
5. Algorithms and Flowcharts
6. Testing and Test Cases
7. User Guide
8. Health Standards Used
9. Checking the Calculations on Real Data
10. Accessibility
11. References

1. INTRODUCTION

Bloom is a desktop application for women's health. It brings together the things a woman may want to track in one private place: body mass index (BMI) and weight, menstrual periods and ovulation, basal body temperature (BBT), mood and symptoms, vital signs and sleep, physical activity, and nutrition.

Many period and fitness apps keep data online, show advertisements, or give numbers without saying where they come from. Bloom keeps every account's data on the user's own computer, protects each account with a hashed password, and bases every calculation on a published health guideline (CDC, WHO, FIGO, Office on Women's Health, ACC/AHA, FDA). The sources are listed inside the app, and the cycle predictions were tested on 1,665 real cycles. Daily logging is rewarded with a flower that grows a petal for each healthy habit.

> **Disclaimer.** Bloom gives general wellness information and estimates. It is not a medical device, it does not diagnose, and its fertile-window predictions are not a method of birth control.

2. OBJECTIVES, SCOPE AND LIMITATIONS

2.1 General objective

To design and build a working, well-tested desktop program that helps women track and understand their health using correct, evidence-based calculations.

2.2 Specific objectives

1. To compute BMI and classify it with the CDC and Asia-Pacific (WHO Western Pacific Region) cut-offs for adults and the CDC growth charts for girls under 20, together with the healthy weight range.
2. To predict the next period, ovulation and fertile window from the user's own cycle history, and to sharpen the prediction once a temperature shift or LH test shows ovulation.
3. To confirm ovulation from basal body temperature using the 3-over-6 rule.
4. To classify blood pressure, heart rate, blood oxygen and sleep, and to track activity and nutrition against WHO, CDC, FDA and NIH guidelines for the user's age.
5. To protect each account with salted, hashed passwords, a lockout after repeated failures, and a one-time recovery code.
6. To validate every input and to prove the logic correct with automated tests, including edge cases.
7. To be usable by people with different abilities: low vision, colour blindness, dyslexia, motion sensitivity, and without a mouse.

2.3 Scope

- Several local accounts on one computer; data saved as JSON files in the `data/` folder.
- Screens: Log in, Sign up and password reset, First-run setup, Today, My Garden, Cycle, Temperature, Journal, Vitals & Sleep, Activity, Nutrition, BMI and Settings.
- Metric and US units, light and dark modes, 6 colour themes, 10 fonts, sound effects, accessibility settings, CSV export and a printable doctor summary.

2.4 Limitations

- Predictions are statistical estimates; irregular cycles lower their accuracy (the app shows a confidence level).
- BMI does not measure body fat, and its categories do not apply during pregnancy.
- Reading aloud is built into the app because Tkinter windows cannot talk to screen readers such as Narrator.
- The app runs offline on one computer; there is no cloud sync between devices.
- Readings are typed by the user; the app does not connect to devices such as thermometers or watches.

3. REQUIREMENTS AND SYSTEM SPECIFICATIONS

3.1 Hardware and software

Item	Requirement
Operating system	Windows 10 or 11 (also runs on macOS with Python 3.13)
Python	3.14 (with Tk 8.6)
Screen	At least 860 × 560 pixels (the window's minimum size)
Libraries	**customtkinter 6.0.0** (widgets), **pillow 13.0.0** (icons, charts, 3D images), **numpy 2.5.4** (3D ray marcher, gradients)
Bundled fonts	Poppins, Nunito, Quicksand, Outfit, Figtree, Fraunces, DM Sans, Atkinson Hyperlegible, Lexend and OpenDyslexic (all under the SIL Open Font License)
Development tools	**pytest 9.2.0** (tests), **ruff 0.17.3** (formatting and style checks)

3.2 Installing and running

```
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
python main.py                      start the app
python -m tracker.ui.preview        open a demo account (demo / bloom2026)
pytest                              run the 763 tests
```

3.3 Functional requirements

No.	Requirement	Where it is met
FR1	Create an account, log in, log out, reset a forgotten password	auth.py, auth_pages.py
FR2	Lock an account for 30 s after 5 wrong attempts	auth.check_login
FR3	Compute BMI, its category and healthy weight range, by percentile under 20	bmi.py, growth.py, bmi_page.py
FR4	Log periods; predict next period, ovulation and fertile window	cycle.py, cycle_page.py
FR5	Show a coloured cycle calendar	calendar_logic.py
FR6	Log BBT and confirm ovulation	bbt.py, bbt_page.py
FR7	Log mood, energy, stress and symptoms	journal.py, journal_page.py
FR8	Classify blood pressure, heart rate, SpO2 and sleep	vitals.py, vitals_page.py

FR9	Track workouts, steps and weekly active minutes	activity.py, activity_page.py
FR10	Track water, meals, caffeine, alcohol, fruit and vegetables	nutrition.py, profile.py
FR11	Give personal insights and a daily checklist	insights.py, home_page.py
FR12	Validate every input with a clear message	validators.py
FR13	Save data safely and survive a damaged file	storage.py
FR14	Reward daily habits with a flower, a garden and achievements	garden.py, garden_page.py
FR15	Offer accessibility settings and full keyboard use	theme.py, widgets.py, speech.py, palette.py, settings_page.py
FR16	Play sound effects made from math, with no audio files	sound.py, sfx.py
FR17	Create a printable summary for a doctor	report.py

3.4 Non-functional requirements

- **Security:** passwords and recovery codes are stored only as PBKDF2-SHA256 hashes (600,000 iterations, 16-byte random salt), as advised by OWASP.
- **Reliability:** files are written to a temporary file and then swapped in, so a crash cannot leave a half-written file.
- **Usability:** errors appear next to the field that caused them; logging takes one or two clicks; text contrast is at least 4.5:1 and 7:1 in high contrast, checked by automated tests.
- **Maintainability:** the logic is separate from the screens, and every public function has a docstring.

4. PROGRAM STRUCTURE AND CODE ORGANIZATION

The program is split into two layers. The **logic layer** (`tracker/`) contains all calculations and rules in plain Python. The **user interface layer** (`tracker/ui/`) builds the windows and calls the logic. This separation means every calculation can be tested and flowcharted on its own, without opening a window.

4.1 Size of the program

Part	Files	Lines of code*	Total lines
Logic (`tracker/`)	21	2,496	3,774
User interface (`tracker/ui/`)	26	10,264	12,151
`main.py`	1	5	12
Tests (`tests/`)	11	1,097	1,530

** Lines of code exclude blank lines, comments and docstrings.*

4.2 Logic modules (tracker/)

Module	Purpose	Main functions
auth.py	Accounts and security	register_user, check_login, reset_password_with_code, change_password, delete_account
storage.py	The only module that reads and writes files	load_users, save_users, load_user_data, save_user_data, export_csv
validators.py	Turns typed text into checked values	parse_number, parse_weight, parse_height, parse_temperature, parse_date, parse_birth_date
units.py	kg/lb, cm/ft-in and °C/°F conversion	kg_to_lb, ft_in_to_cm, c_to_f, format_weight
profile.py	Age, calorie target, water goal, goals	bmr_mifflin_st_jeor, estimated_energy_requirement_girl, calorie_target, suggested_water_ml
bmi.py	BMI and adult categories	calculate_bmi, classify_bmi, healthy_weight_range, calculate_age
growth.py	BMI percentiles for girls under 20 (CDC growth charts)	assess, percentile, z_score
cycle.py	Periods, cycle statistics and predictions	add_period, cycle_lengths, predict_cycle, with_ovulation_sign, phase_on, phase_segments

calendar_logic.py	Month grid and day colours	month_grid, day_status
bbt.py	Basal body temperature and ovulation signs	readings_between, detect_thermal_shift, ovulation_sign, usual_days_to_period
daylog.py, journal.py	Daily logs, mood and symptom patterns	peek_day, sleep_mood_pairs
vitals.py	Blood pressure, heart rate, SpO2, sleep	bp_category, bp_average, spo2_note, recommended_sleep, sleep_debt
activity.py	Workouts and active minutes	workout_kcal, intensity, moderate_minutes, weekly_goal_minutes, strength_days
nutrition.py	Water, caffeine, alcohol, food, nutrient needs	expected_water_by_now, caffeine_limit, protein_target_g, daily_nutrients
garden.py	The daily flower, the garden and achievements	petals, next_petal, garden_days, achievement_progress
sound.py	Sound effects made from math	note_samples, render
insights.py	Smart tips and the daily checklist	prediction_for, build_insights, daily_checklist
report.py	The printable doctor summary	doctor_summary_html
demo.py	90 days of made-up demo data	make_demo_data

4.3 User interface modules (tracker/ui/)

Module	Purpose
app.py	The main window (class **App**, a subclass of `ctk.CTk`), the sidebar, page switching, saving and keyboard shortcuts
palette.py	Quick Find (Ctrl+K) and the keyboard shortcuts list (F1)
page.py	Base class **Page**; every screen inherits from it and overrides `build()`
auth_pages.py, onboarding.py	Log in, sign up, password reset, recovery code, and the 7-step setup
home_page.py … settings_page.py	One file per sidebar page (10 pages)
widgets.py	Reusable pieces: cards, buttons, inputs, charts, the flower and garden, keyboard access, dialogs, toasts
theme.py, anim.py, effects.py	Colours, accents, high contrast and colour-blind palettes, fonts, spacing; animation easing and timers; confetti, card glow and page transitions

sfx.py, speech.py	Playing the sound effects; reading aloud with the computer's own voice
icons.py, graphics.py, render3d.py	Icons, charts and 3D shapes drawn with Pillow and numpy
preview.py	Opens the app with a demo account in a temporary folder

4.4 Design rules

1. **Logic never imports the user interface.** Modules in `tracker/` do not import customtkinter, tkinter or Pillow.
2. **Every public function has a docstring** that says what it returns and which errors it raises.
3. **Only storage.py touches files.** Inside the program, dates are `datetime.date` objects; in the files they are ISO text such as `"2026-10-31"`.

Rules 1 and 2 are checked automatically by `tests/test_architecture.py`.

4.5 Programming constructs used

- **Selection:** if / elif / else chains (e.g. `bp_category` checks from the most serious category down).
- **Repetition:** `for` loops over tables and lists (BMI categories, cycle lengths, the 3-over-6 window, the last 7 days); nested loops build the calendar grid.
- **Functions** with type hints, default arguments and return values; **constants** in capitals instead of "magic numbers".
- **Data structures:** lists, dictionaries, tuples and frozen `dataclass` objects (`CyclePrediction`, `ThermalShift`).
- **Exception handling:** the logic raises `ValueError` or `AuthError`; the screens catch them with try/except and show the message.
- **Classes and inheritance:** `App` inherits from `ctk.CTk`; every page inherits from `Page`.

4.6 Data storage

`data/users.json` holds the accounts (username, password hash, salt, iterations, recovery-code hash, failed attempts and lock time). Each user also has a private data file named after the user id (e.g. `u_3f9a1c2b`) holding the profile, goals, settings, periods, weights and daily logs. Example of a period and a day log:

```
"periods": [{"start": "2026-10-20", "end": "2026-10-24"}],
"days": {"2026-11-03": {"bbt_c": 36.62, "mood": 4, "sleep_h": 7.5,
                        "water_ml": 750, "bp_sys": 112, "bp_dia": 74}}
```

5. ALGORITHMS AND FLOWCHARTS

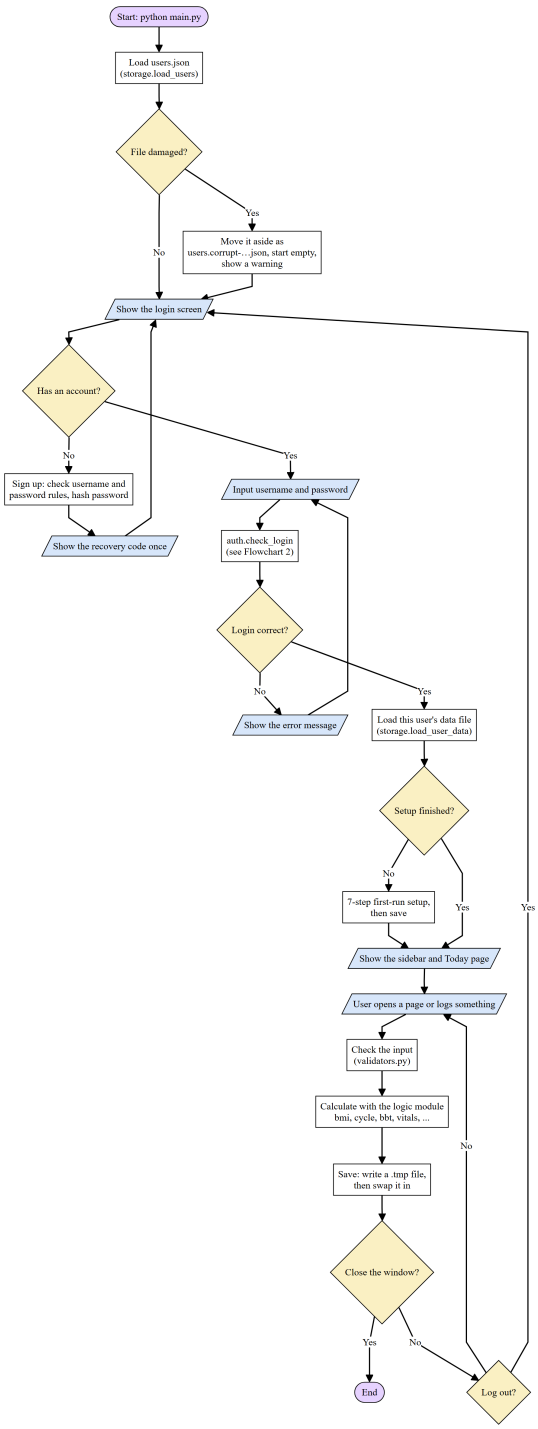
Each flowchart follows its function's code step by step, so it can be compared with the code directly. Symbols: **oval** = start/end, **parallelogram** = input/output, **rectangle** = process, **diamond** = decision.

5.1 The whole program

The program starts in **main.py**, which creates the **App** window. App.start() loads the accounts; a damaged file is moved aside, not deleted. The user signs up or logs in, finishes the setup once, and then repeats the main loop: open a page or log something, check the input, calculate, and save. Logging out returns to the login screen; closing the window ends the program.

Flowchart 1. The Whole Program

Code: tracker/ui/app.py (App.start, log_in, enter_main, show_page)

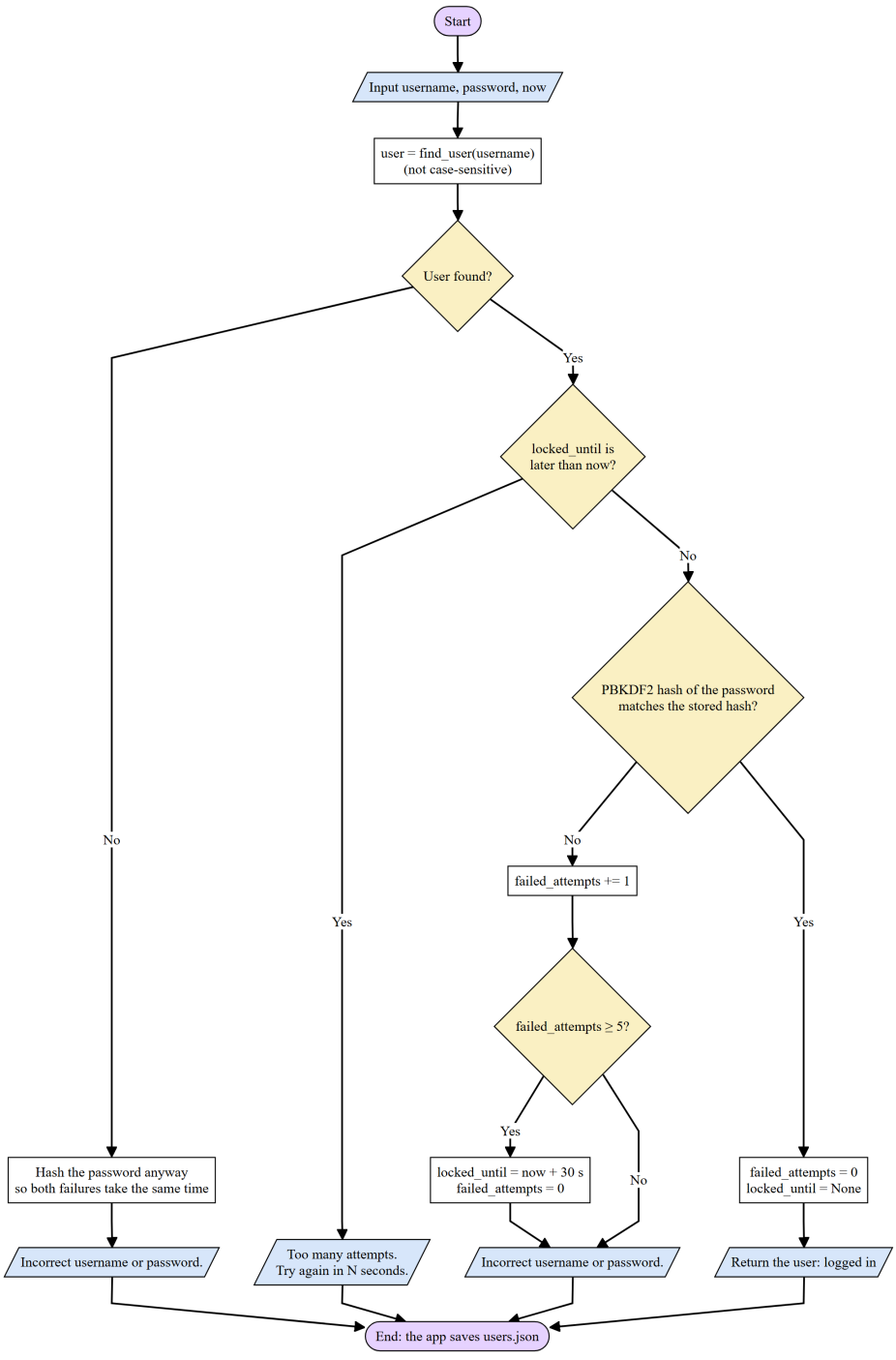


5.2 Logging in with the 5-try lockout

check_login() hashes the password even when the username does not exist, so an attacker cannot tell from the timing which part was wrong. Both failures show the same message. Every wrong password adds 1 to failed_attempts; the 5th locks the account for 30 seconds and resets the counter.

Flowchart 2. Logging In with the 5-Try Lockout

Code: auth.check_login

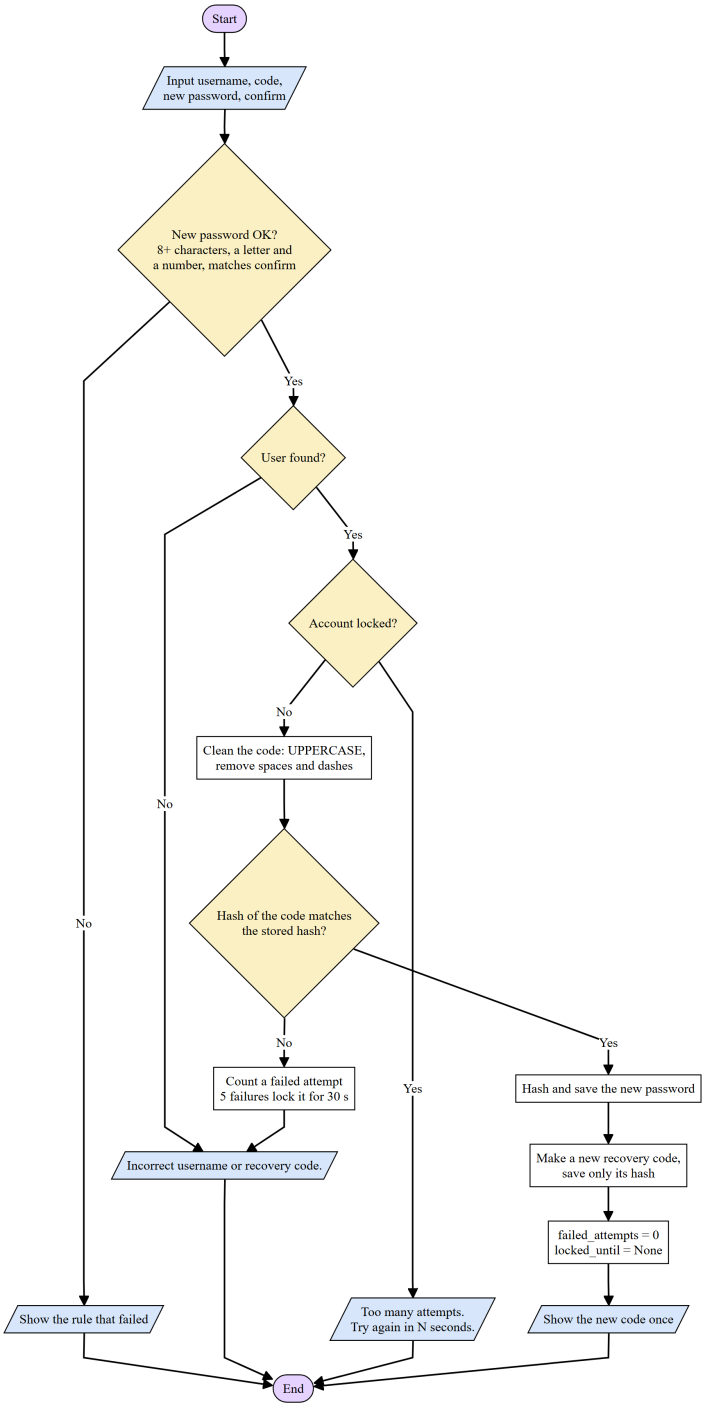


5.3 Resetting a password with the recovery code

The new password's rules are checked *first*, so a typing mistake there does not count as a failed attempt. The code is cleaned (uppercase, no spaces or dashes) before hashing. After a successful reset a **new** code replaces the old one, so each code works only once.

Flowchart 3. Resetting a Password with the Recovery Code

Code: auth.reset_password_with_code

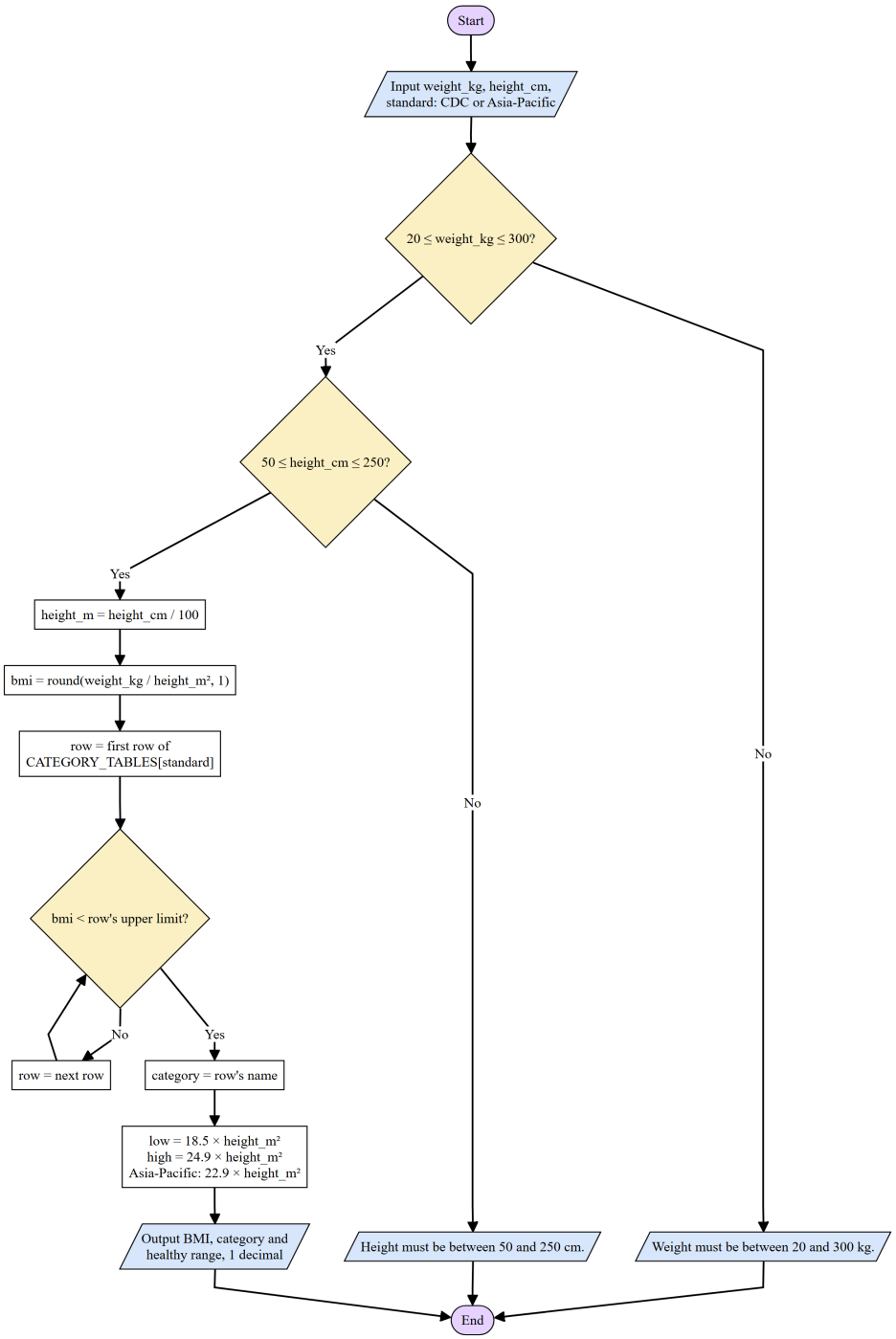


5.4 BMI, its category and the healthy range

The inputs are range-checked, the BMI is rounded to 1 decimal, and the category comes from a **loop over a table** instead of a chain of if-statements. The rounded value is classified so that the label always matches the number on screen (TC02–TC03).

Flowchart 4. BMI, Its Category and the Healthy Range

Code: bmi.calculate_bmi, classify_bmi, healthy_weight_range

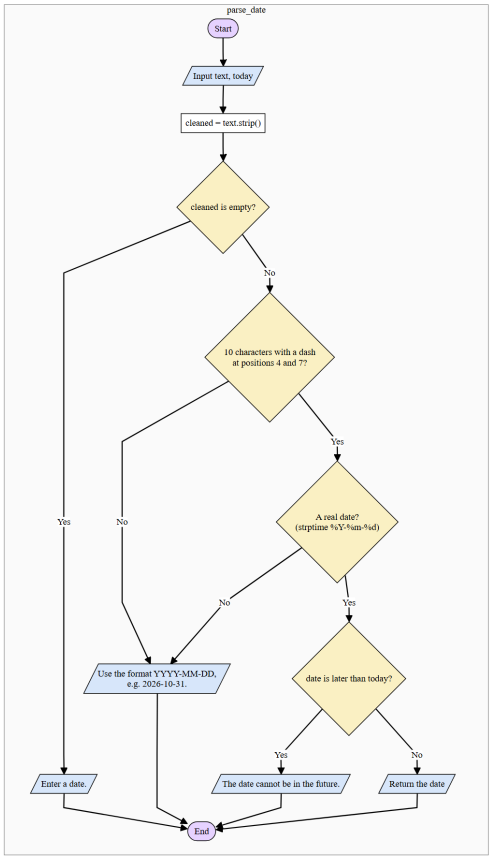


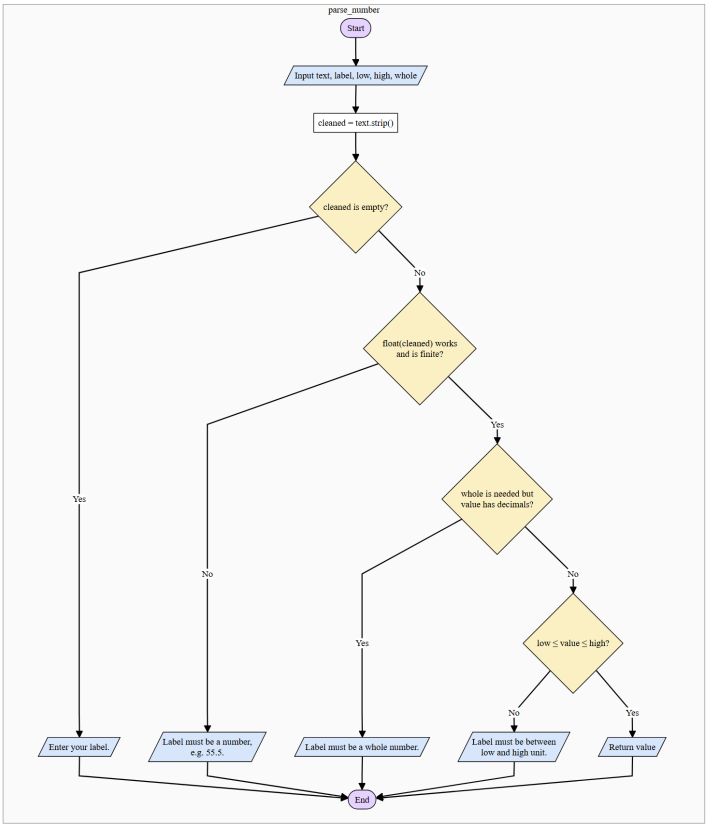
5.5 Checking typed numbers and dates

Every value typed by the user passes through validators.py. Each check that fails raises a ValueError with a message that says what is wrong and how to fix it; the screen shows that message next to the field.

Flowchart 5. Checking Typed Numbers and Dates

Code: validators.parse_number, parse_date



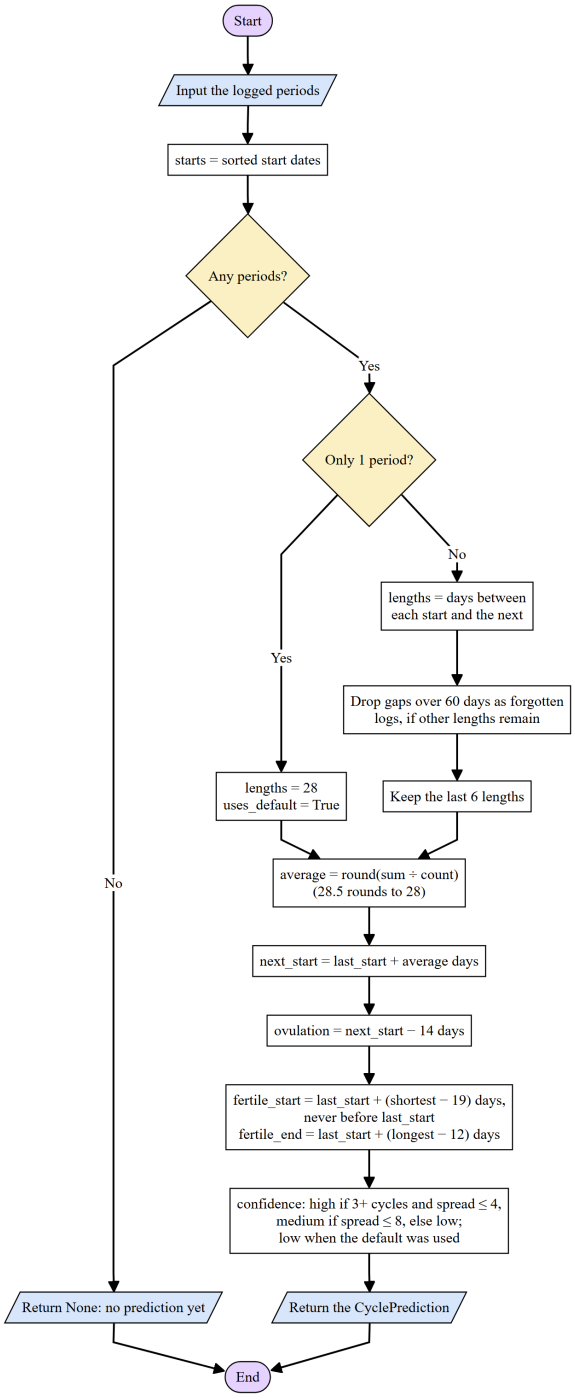


5.6 Predicting the next period and fertile window

The prediction averages the last 6 cycle lengths, ignoring gaps over 60 days (a forgotten log). Ovulation is counted **back 14 days** from the next period because the luteal phase is about 14 days long. The fertile window uses the calendar method: shortest cycle − 18 to longest cycle − 11 (the extra − 1 is because day 1 is the start date itself), widened while few cycles are known. Section 9 shows the tests behind each choice.

Flowchart 6. Predicting the Next Period and Fertile Window

Code: cycle.predict_cycle

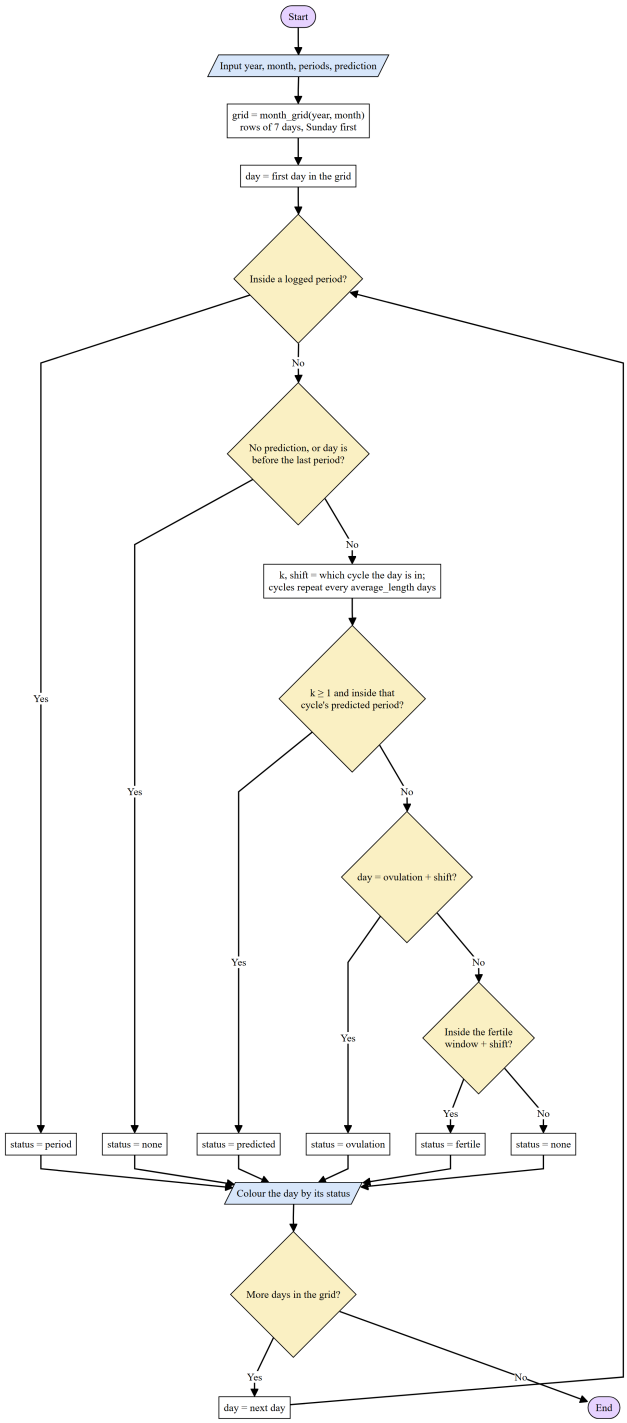


5.7 The month grid and each day's colour

The grid comes from Python's calendar module. Each day is tested in a fixed order: logged period, predicted period, ovulation, fertile window. The first match decides the colour. Predictions repeat every average cycle length, so future months are coloured too.

Flowchart 7. The Month Grid and Each Day's Colour

Code: calendar_logic.month_grid, day_status

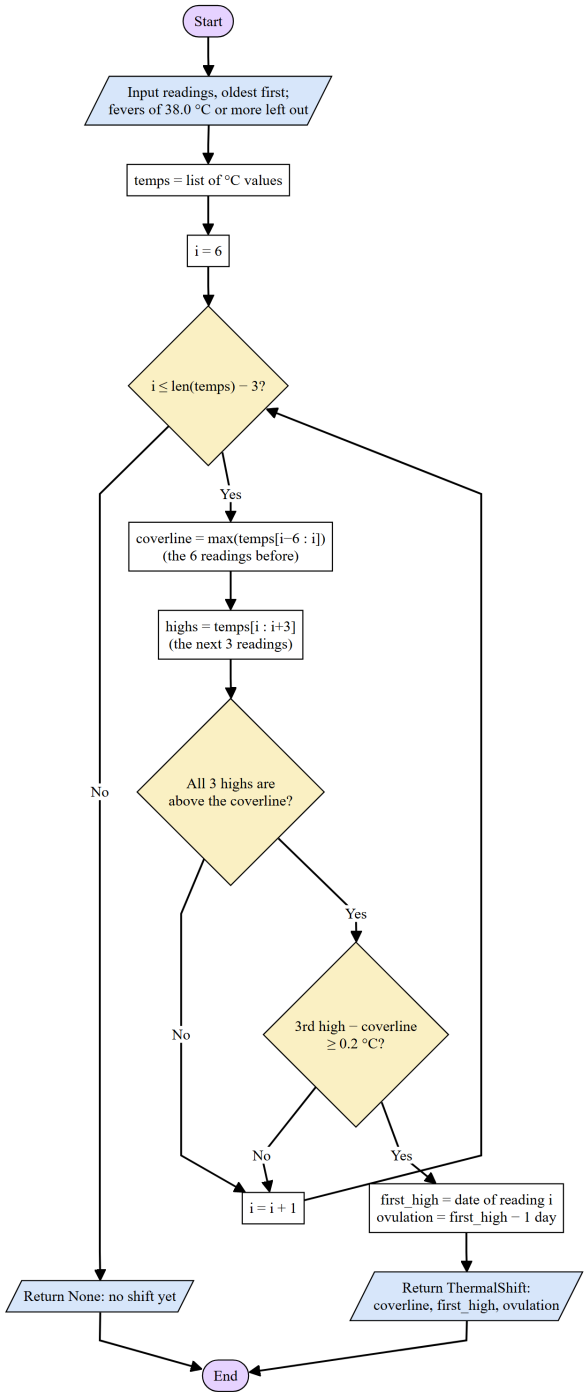


5.8 Confirming ovulation with the 3-over-6 rule

A loop slides along the readings. The coverline is the highest of the 6 readings before position i; the next 3 readings must all be above it, and the 3rd must be at least 0.2 °C higher, or a 4th reading must also be above the coverline. Fevers (38.0 °C or more) and readings the user marks as disturbed are removed first, because illness, alcohol or a late night, not progesterone, raised them.

Flowchart 8. Confirming Ovulation with the 3-over-6 Rule

Code: bbt.detect_thermal_shift

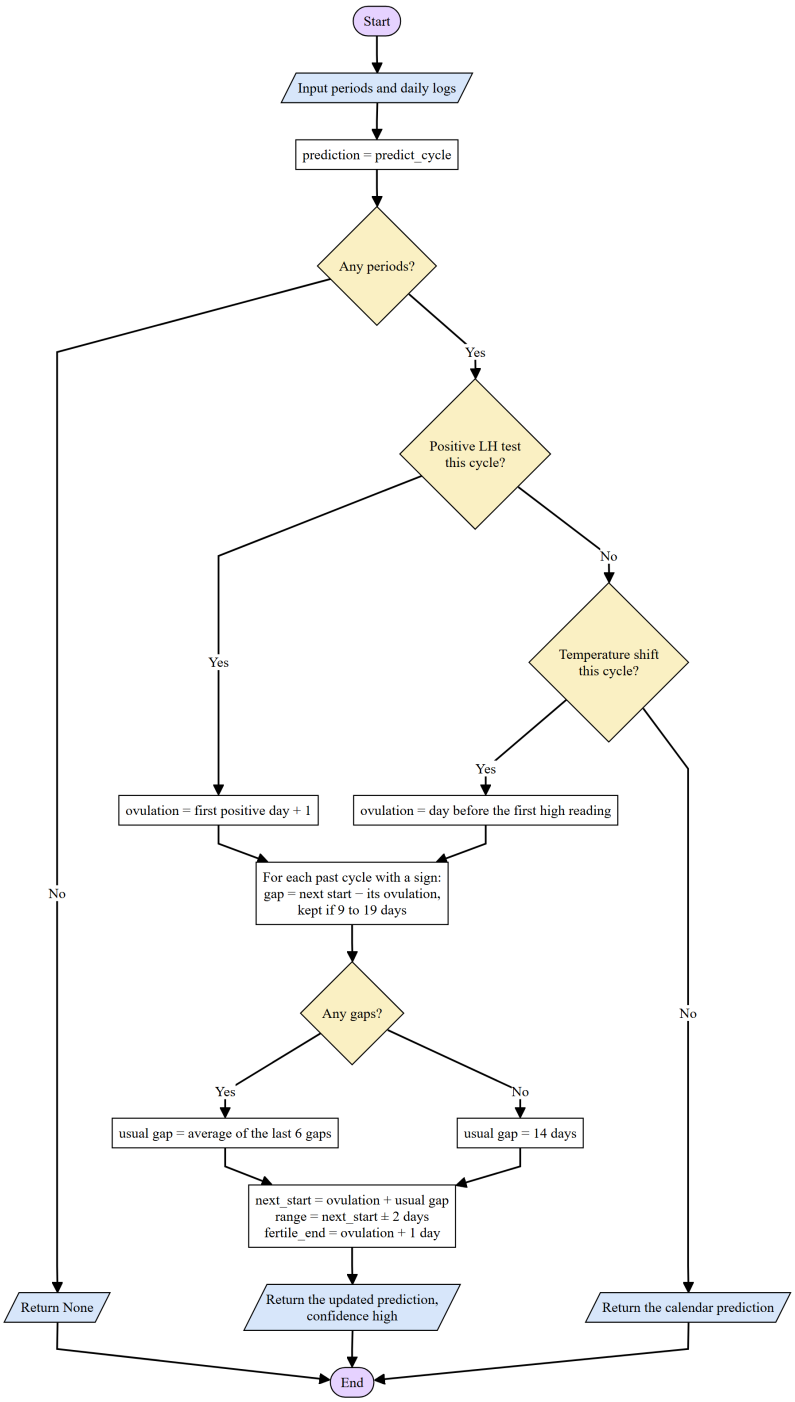


5.9 Counting the next period from a known ovulation

The luteal phase varies less than the follicular phase, so once a positive LH test or a temperature shift shows ovulation, the next period is counted from it. Past cycles with a sign give the person's own usual gap. On real data this cut the average error from 2.0 to 1.2 days (Section 9).

Flowchart 9. Counting the Next Period from a Known Ovulation

Code: insights.prediction_for, bbt.ovulation_sign, usual_days_to_period, cycle.with_ovulation_sign

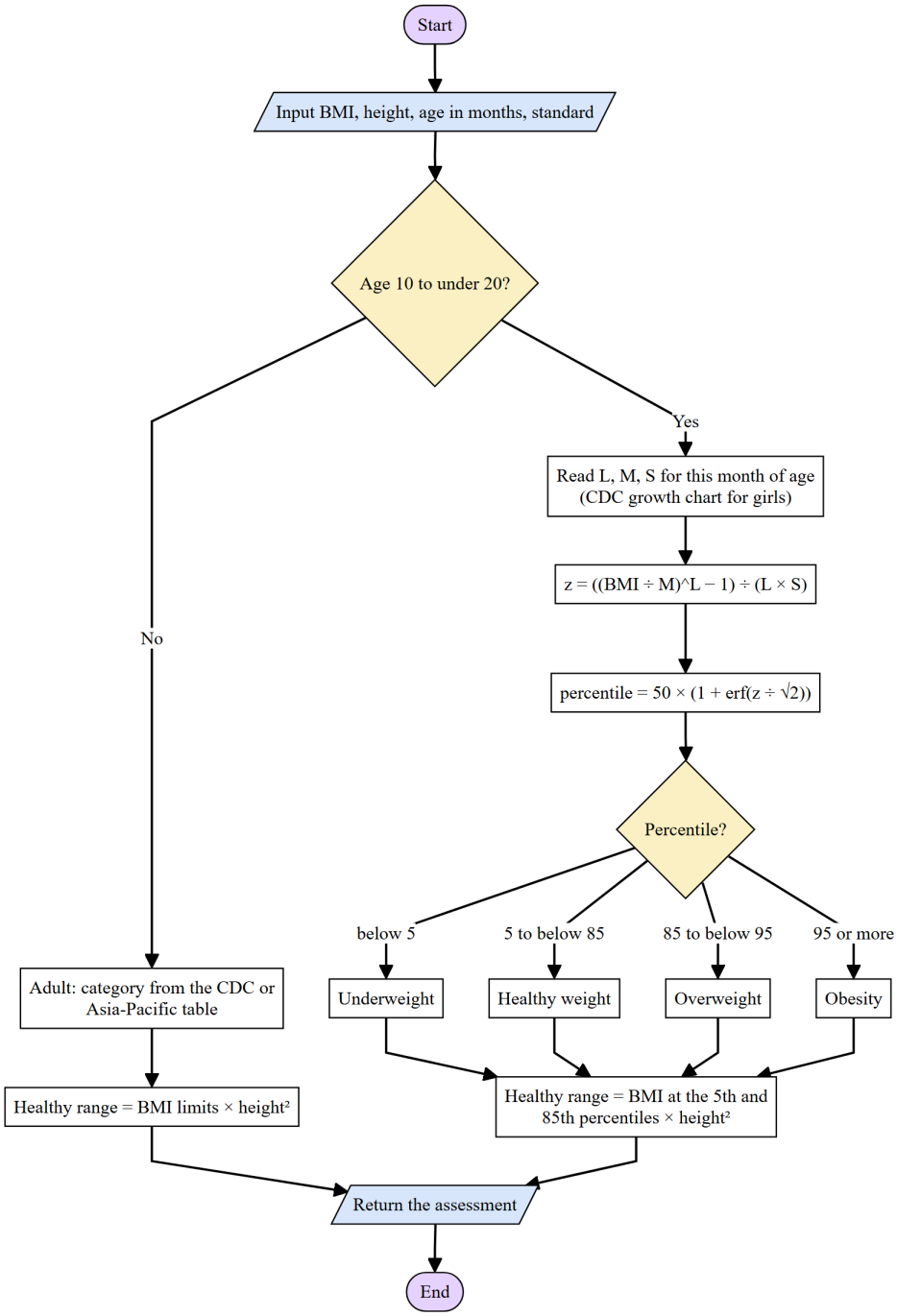


5.10 What a BMI means under and over 20

Adult categories do not fit growing bodies. From 10 to under 20, the BMI is turned into a z-score with the CDC's L, M and S values for that month of age, then into a percentile, and the category comes from the percentile. The same BMI can therefore mean different things at different ages (see the test cases).

Flowchart 10. What a BMI Means Under and Over 20

Code: growth.assess, percentile

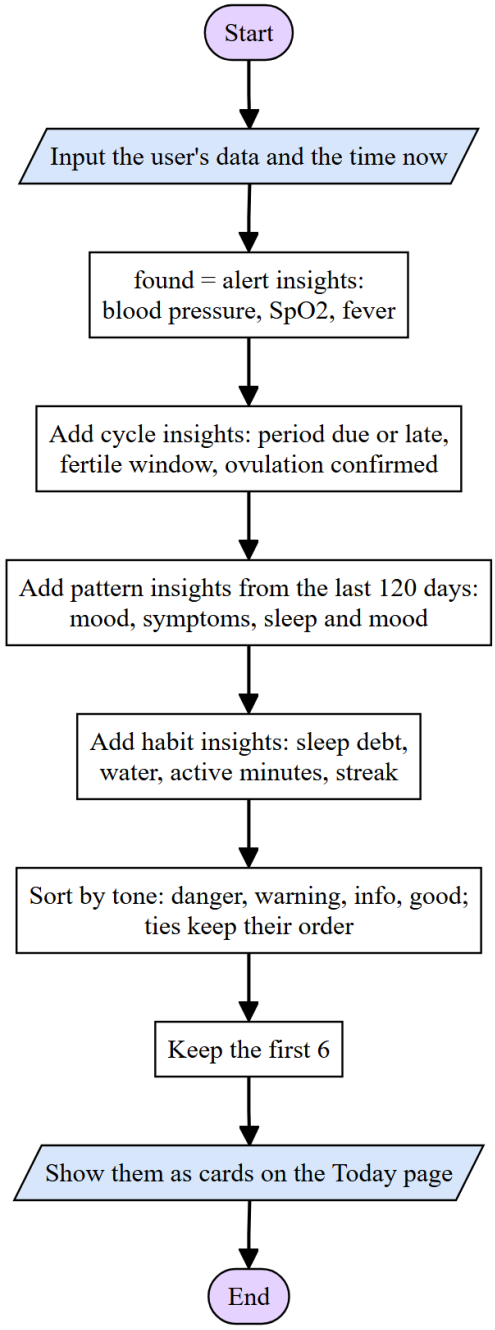


5.11 Choosing the insights for the Today page

Five groups of rules (alerts, cycle, body, patterns, habits) each add their insights to one list. The list is sorted by tone so urgent alerts come first (Python's sort is stable, so ties keep their order), and only the first 6 are shown.

Flowchart 11. Choosing the Insights for the Today Page

Code: insights.build_insights



6. TESTING AND TEST CASES

6.1 Automated unit tests

The logic is tested with **pytest**. There are **763 tests** in 11 files (test_accuracy, test_architecture, test_auth, test_bmi, test_cycle, test_garden_sound, test_health, test_insights, test_look, test_storage, test_validators). test_look checks the contrast of every colour pair in every colour theme, which is why the number is large. The storage tests use a temporary folder, so real data is never touched. The latest run:

```
$ pytest
763 passed
```

6.2 Test case table

The table below lists 69 representative test cases, including edge cases (marked *(edge)*). The **actual output** column was produced by running each input through the real functions of the program, not typed by hand. **69 of 69 passed.** Dates assume today is 2026-11-03.

No.	Module	Test case	Input	Expected output	Actual output	Result
TC01	BMI	Normal BMI	55 kg, 160 cm	21.5	21.5	**Pass**
TC02	BMI	rounds up to the limit *(edge)*	47.36 kg, 160 cm (raw 18.4999)	18.5	18.5	**Pass**
TC03	BMI	label matches rounded BMI *(edge)*	BMI 18.5, CDC	Healthy weight	Healthy weight	**Pass**
TC04	BMI	CDC overweight	BMI 27.0, CDC	Overweight	Overweight	**Pass**
TC05	BMI	Asia-Pacific lower cut-off	BMI 23.0, Asia-Pacific	Overweight	Overweight	**Pass**
TC06	BMI	Asia-Pacific obesity class I	BMI 26.0, Asia-Pacific	Obesity class I	Obesity class I	**Pass**
TC07	BMI	Healthy weight range	160 cm, CDC	(47.4, 63.7)	(47.4, 63.7)	**Pass**
TC08	BMI	Invalid: weight too high	350 kg, 160 cm	Error: Weight must be between 20 and 300 kg.	Error: Weight must be between 20 and 300 kg.	**Pass**
TC09	BMI	birthday not yet reached *(edge)*	born 2004-12-01, today 2026-11-03	21	21	**Pass**
TC10	BMI	adult categories at 19 *(edge)*	age 19	False	False	**Pass**
TC11	Input	Text instead of a number	weight "abc"	Error: Weight must be a number, e.g. 55.5.	Error: Weight must be a number, e.g. 55.5.	**Pass**

No.	Module	Test case	Input	Expected output	Actual output	Result
TC12	Input	Blank field	weight ""	Error: Enter your weight.	Error: Enter your weight.	**Pass**
TC13	Input	Height out of range	height "300"	Error: Height must be between 50 and 250 cm.	Error: Height must be between 50 and 250 cm.	**Pass**
TC14	Input	height typed in metres *(edge)*	height "1.6"	Error: Height must be between 50 and 250 cm. Did you enter metres? Type 160 for 1.60 m.	Error: Height must be between 50 and 250 cm. Did you enter metres? Type 160 for 1.60 m.	**Pass**
TC15	Input	Pounds converted to kg	"121" lb	54.9	54.9	**Pass**
TC16	Input	impossible date *(edge)*	"2026-02-30"	Error: Use the format YYYY-MM-DD, e.g. 2026-10-31.	Error: Use the format YYYY-MM-DD, e.g. 2026-10-31.	**Pass**
TC17	Input	Future date	"2026-12-25"	Error: The date cannot be in the future.	Error: The date cannot be in the future.	**Pass**
TC18	Input	"nan" is not a number *(edge)*	weight "nan"	Error: Weight must be a number, e.g. 55.5.	Error: Weight must be a number, e.g. 55.5.	**Pass**
TC19	Cycle	Cycle lengths	starts Jan 3, Jan 28, Mar 4, Apr 3	[25, 35, 30]	[25, 35, 30]	**Pass**
TC20	Cycle	Next period	starts Jan 1, Jan 29, Feb 26 (28-day cycles)	2026-03-26	2026-03-26	**Pass**
TC21	Cycle	Ovulation = next period - 14	same	2026-03-12	2026-03-12	**Pass**
TC22	Cycle	Fertile window: 2 known cycles, 1-day margin	same	2026-03-06 to 2026-03-15	2026-03-06 to 2026-03-15	**Pass**
TC23	Cycle	Likely range for the next period	same	2026-03-24 to 2026-03-28	2026-03-24 to 2026-03-28	**Pass**
TC24	Cycle	Fertile window: 4 known cycles, no margin	4 cycles of 28 days	2026-05-02 to 2026-05-09	2026-05-02 to 2026-05-09	**Pass**
TC25	Cycle	Next period from a known ovulation	ovulation Mar 15, usual gap 12 days	2026-03-27	2026-03-27	**Pass**
TC26	Cycle	late period moves up to today *(edge)*	next period Mar 26, today Apr 1	2026-04-01	2026-04-01	**Pass**

No.	Module	Test case	Input	Expected output	Actual output	Result
TC27	Cycle	phase unknown after a long gap *(edge)*	last period Feb 26, today Nov 3	None	None	**Pass**
TC28	Cycle	Teen cycles: wider typical range	[22, 44], age 16	[]	[]	**Pass**
TC29	Cycle	only one period logged *(edge)*	start Oct 1	28 days, default used	28 days, default used	**Pass**
TC30	Cycle	no periods *(edge)*	no periods	None	None	**Pass**
TC31	Cycle	x.5 average rounds to even *(edge)*	cycles 28 and 29 (average 28.5)	28	28	**Pass**
TC32	Cycle	Confidence with steady cycles	4 cycles of 28 days	high	high	**Pass**
TC33	Cycle	Lengths outside 24-38 days	[22, 28, 40]	[22, 40]	[22, 40]	**Pass**
TC34	Cycle	Duplicate period	Oct 1 logged twice	Error: This period is already logged.	Error: This period is already logged.	**Pass**
TC35	Cycle	period over 15 days *(edge)*	Oct 1 to Oct 20	Error: A period cannot last more than 15 days.	Error: A period cannot last more than 15 days.	**Pass**
TC36	Cycle	Cycle day	period Oct 20, day Nov 3	15	15	**Pass**
TC37	BBT	3-over-6 shift found	6 lows (max 36.5), then 36.7, 36.8, 36.8	coverline 36.5, ovulation 2026-10-06	coverline 36.5, ovulation 2026-10-06	**Pass**
TC38	BBT	rise under 0.2 C *(edge)*	6 lows (max 36.5), then 36.6 x3	None	None	**Pass**
TC39	BBT	a 4th reading confirms a small rise *(edge)*	6 lows (max 36.5), then 36.6 x4	first high 2026-10-07	first high 2026-10-07	**Pass**
TC40	BBT	fever left out *(edge)*	38.0 C	True	True	**Pass**
TC41	BBT	A reading marked as disturbed is left out	37.2 C marked by the user	[]	[]	**Pass**
TC42	Vitals	Normal blood pressure	115/75	('Normal', 'good')	('Normal', 'good')	**Pass**
TC43	Vitals	diastolic alone raises it *(edge)*	118/92	('Stage 2 hypertension', 'warning')	('Stage 2 hypertension', 'warning')	**Pass**
TC44	Vitals	Severe	185/100	('Severe hypertension', 'danger')	('Severe hypertension', 'danger')	**Pass**

No.	Module	Test case	Input	Expected output	Actual output	Result
TC45	Vitals	low blood pressure *(edge)*	85/70	('Low', 'info')	('Low', 'info')	**Pass**
TC46	Vitals	Oxygen 92% is urgent (NHS)	92%	danger	danger	**Pass**
TC47	Vitals	Sleep recommendation by age	age 15	8.0	8.0	**Pass**
TC48	Vitals	Sleep debt (extra sleep doesn't cancel)	goal 8 h; nights 6, 9, 7	3.0	3.0	**Pass**
TC49	Activity	MET calories	Brisk walking 4.3 MET, 55 kg, 30 min	124	124	**Pass**
TC50	Activity	Vigorous minutes count double	20 min running (9.8 MET) + 30 min walking (3.5)	70	70	**Pass**
TC51	Nutrition	Calorie target (Mifflin-St Jeor)	55 kg, 160 cm, 22 y, lightly active	1760	1760	**Pass**
TC52	Nutrition	Water goal	55 kg	1750	1750	**Pass**
TC53	Nutrition	Calories for a teen (IOM equation)	50 kg, 160 cm, 15 y, moderately active	2310	2310	**Pass**
TC54	Nutrition	Caffeine mark for a teen	age 16	100	100	**Pass**
TC55	Nutrition	Caffeine mark when trying to conceive	age 30, conceiving	200	200	**Pass**
TC56	Nutrition	Protein target	50 kg, adult	40	40	**Pass**
TC57	Nutrition	Iron need by age	age 30	18 mg	18 mg	**Pass**
TC58	Activity	WHO goal under 18	age 16	420	420	**Pass**
TC59	BMI	Under 20: category from the percentile	BMI 26, 15 years	Overweight	Overweight	**Pass**
TC60	BMI	same BMI, adult *(edge)*	BMI 24.5, 25 years	Healthy weight	Healthy weight	**Pass**
TC61	BMI	same BMI, 15 years *(edge)*	BMI 24.5, 15 years	Overweight	Overweight	**Pass**

No.	Module	Test case	Input	Expected output	Actual output	Result
TC62	Garden	A full day earns all 6 petals	steps, water, sleep, mood, vitals and a meal logged	6	6	**Pass**
TC63	Account	Username too short	"ab"	Error: Username must be 3 to 20 characters.	Error: Username must be 3 to 20 characters.	**Pass**
TC64	Account	Password without a number	"password"	Error: Password needs at least one letter and one number.	Error: Password needs at least one letter and one number.	**Pass**
TC65	Account	Correct login	maria / bloom2026	maria	maria	**Pass**
TC66	Account	Wrong password	maria / wrong123	Error: Incorrect username or password.	Error: Incorrect username or password.	**Pass**
TC67	Account	locked after 5 wrong tries *(edge)*	6th try, right password	Error: Too many attempts. Try again in 30 seconds.	Error: Too many attempts. Try again in 30 seconds.	**Pass**
TC68	Account	Unlocked after 30 seconds	same, 31 s later	maria	maria	**Pass**
TC69	Account	Password stored as a hash	stored record	True	True	**Pass**

6.3 System testing

The whole application was also driven from start to finish by a script with 113 checks: creating an account, saving the recovery code and finishing the full setup; opening every page; logging data on each page; every colour theme, font, text size and accessibility setting; Quick Find, dialogs and keyboard focus; pages prepared in the background; logging out and back in; an empty account and a years-old log. A second script walked every page at different fonts, text sizes and window sizes looking for text that did not fit. No errors occurred.

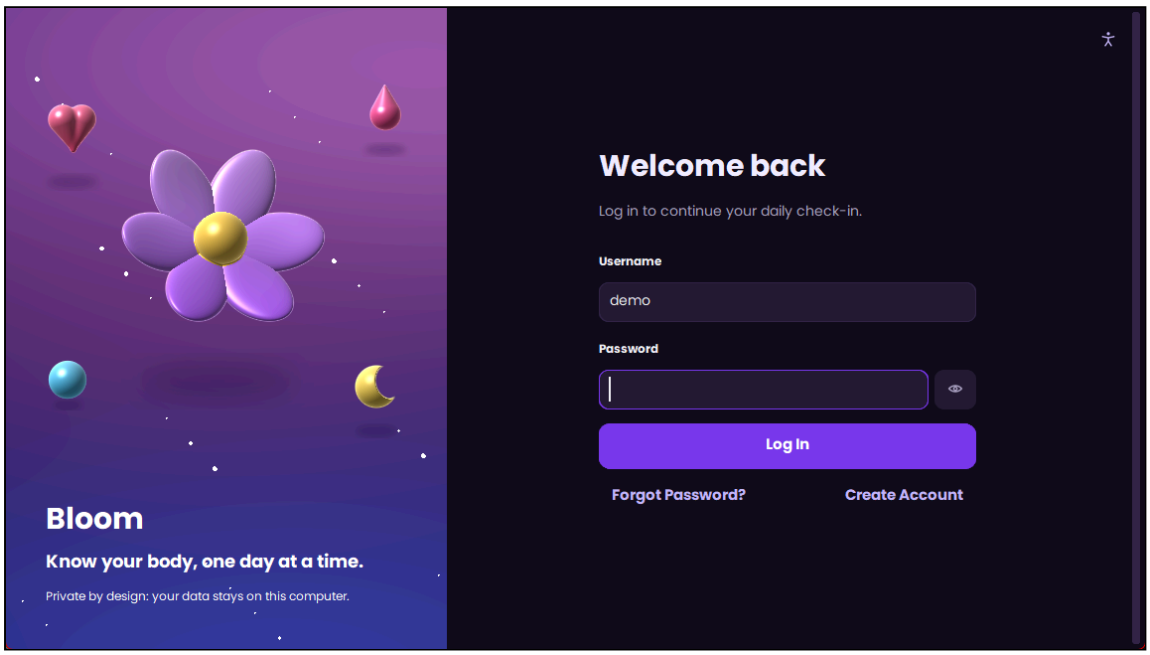
Speed was measured as well. Building a page takes about three seconds on the test laptop, so while the user reads, Bloom prepares the other pages in the background, one at a time, and pauses at any click or key. A prepared page opens in about a third of a second.

7. USER GUIDE

Start the app with `python main.py`. The screenshots below use the demo account (`python -m tracker.ui.preview`, username **demo**, password **bloom2026**) in dark mode. Press **F1** in the app for every keyboard shortcut.

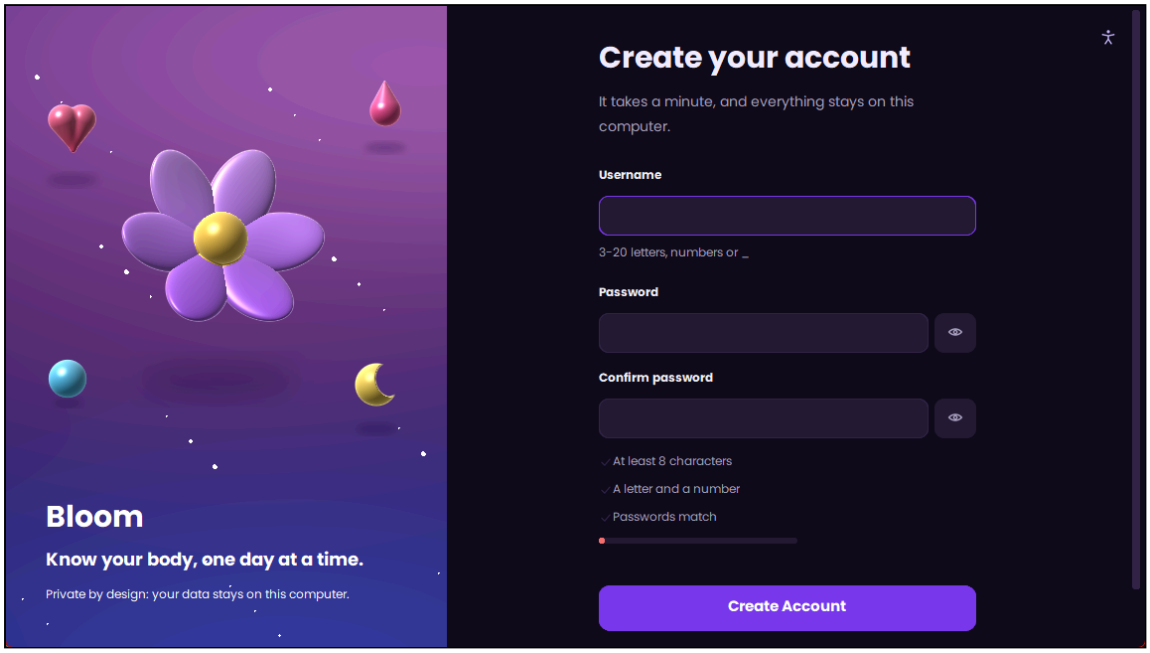
7.1 Log in

Type your username and password and press **Log In** (or Enter). New users click **Create Account**. If you forget your password, click **Forgot Password?** and use your recovery code.



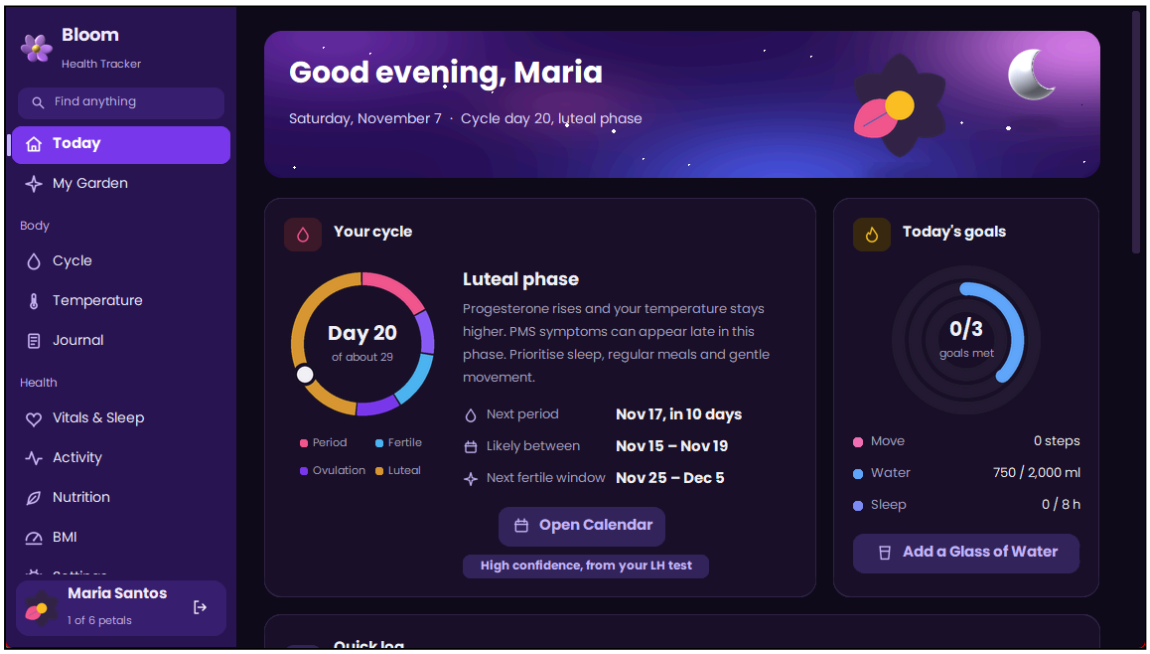
7.2 Create an account

Choose a username (3–20 letters, numbers or _) and a password (8+ characters with a letter and a number). Write down the recovery code shown next; it appears only once. A 7-step setup then asks for your name, birth date, height and weight, cycle, activity level and goals.



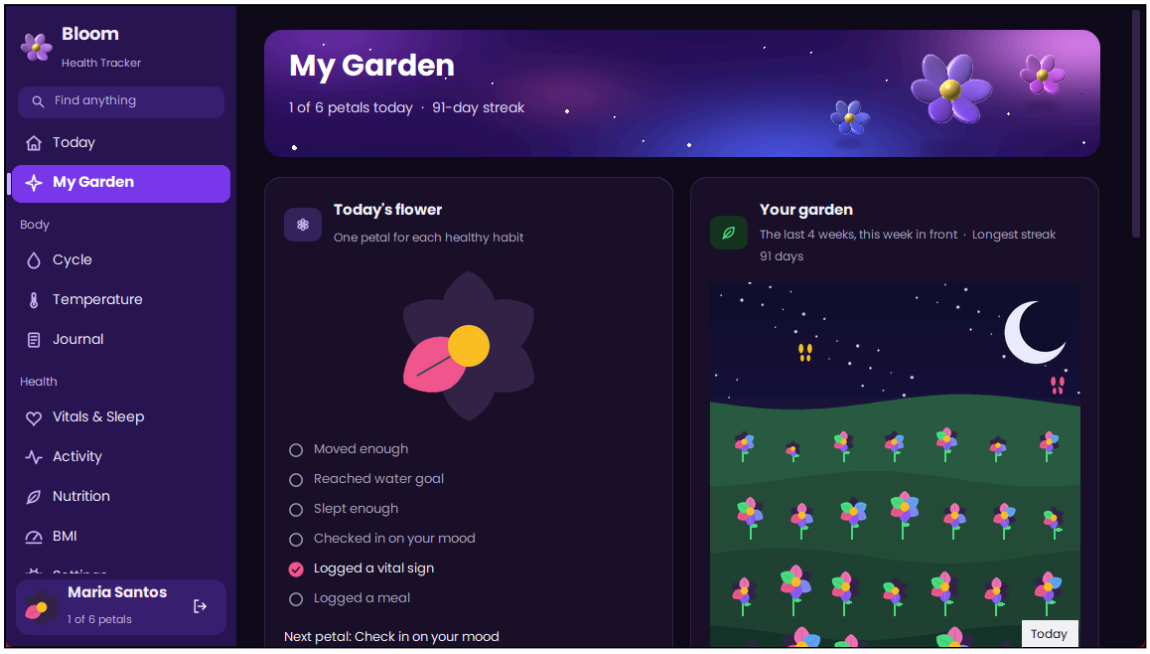
7.3 Today

Shows your greeting under a banner in your theme colour, with the sun by day and the moon at night, today's flower, the cycle wheel with every phase, the likely range of your next period, today's goals, quick logging buttons, smart insights, a daily checklist and your last 7 days. Ctrl+K finds any page, setting or action.



7.4 My Garden

A flower with one petal per healthy habit, each in its own colour, and the next petal to earn. The last 4 weeks grow into a garden, this week in front. Below are 9 achievements with progress bars and a guided 4-4-6 breathing exercise.



7.5 Cycle

Log a period's start and end, flow, cervical mucus and LH tests. The calendar is coloured: logged period, predicted period, fertile window and ovulation. Predictions show a confidence level.



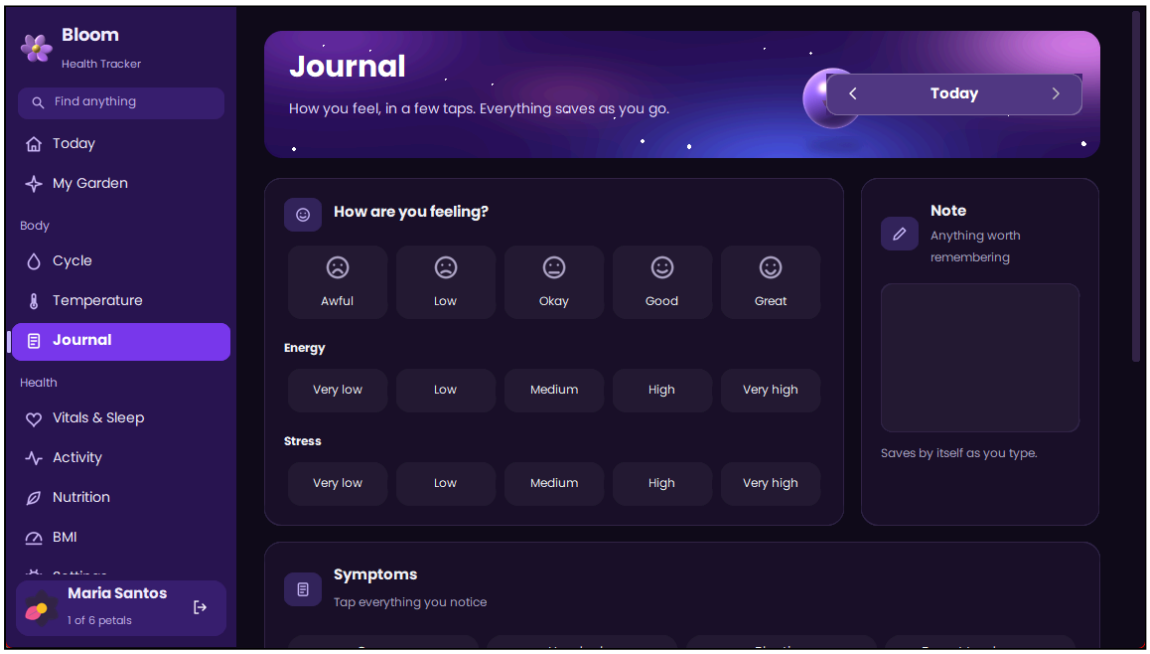
7.6 Temperature (BBT)

Log your temperature each morning before getting up. The chart draws the coverline and marks ovulation once the 3-over-6 rule confirms it. After a cold, alcohol or a late night, choose **Leave this reading out**.



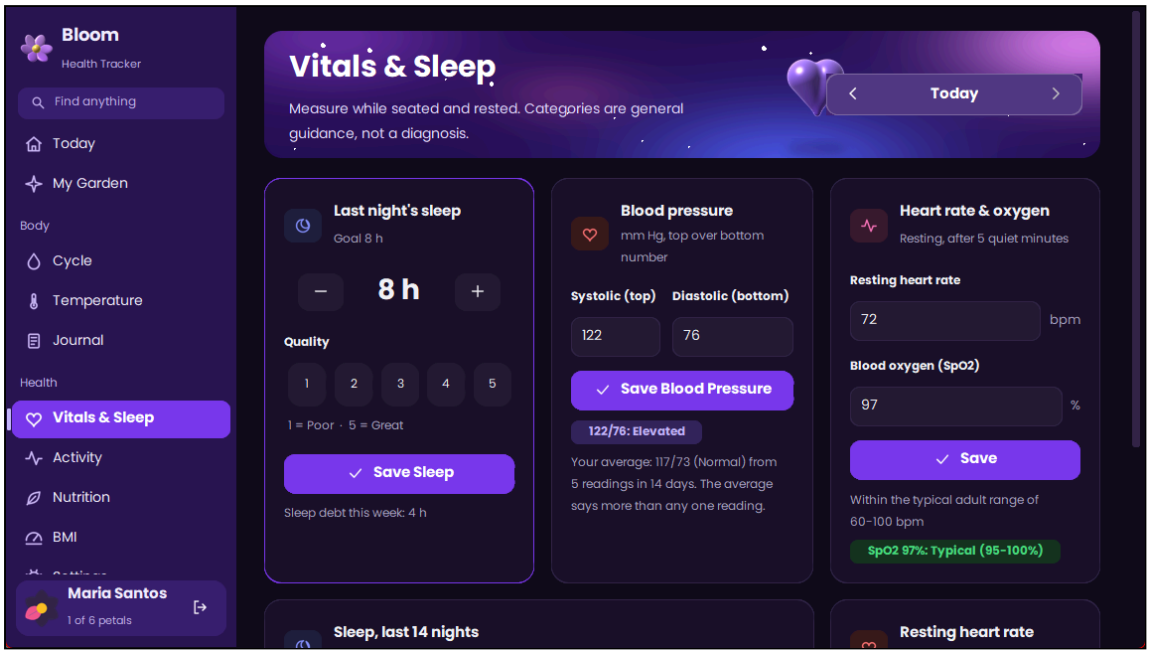
7.7 Journal

Log mood, energy, stress, symptoms and a note for any day. The page shows which symptoms cluster in which phase and a 14-day mood trend.



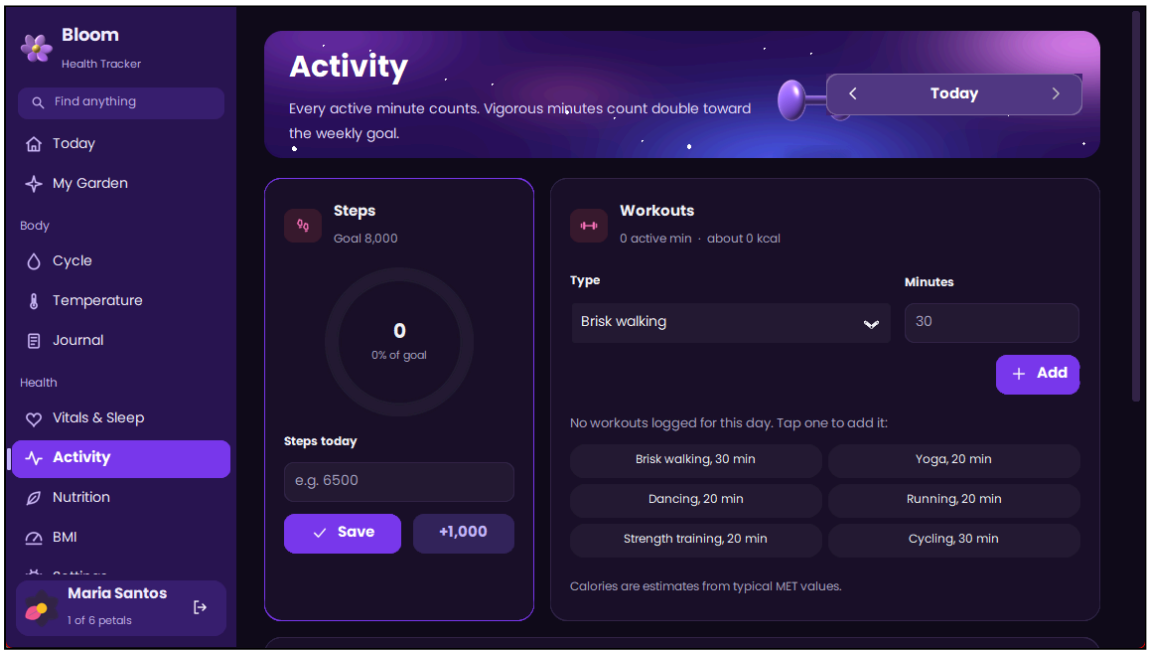
7.8 Vitals & Sleep

Log sleep hours and quality, blood pressure, resting heart rate and blood oxygen. Each reading gets a category and a colour; 14-day charts show trends.



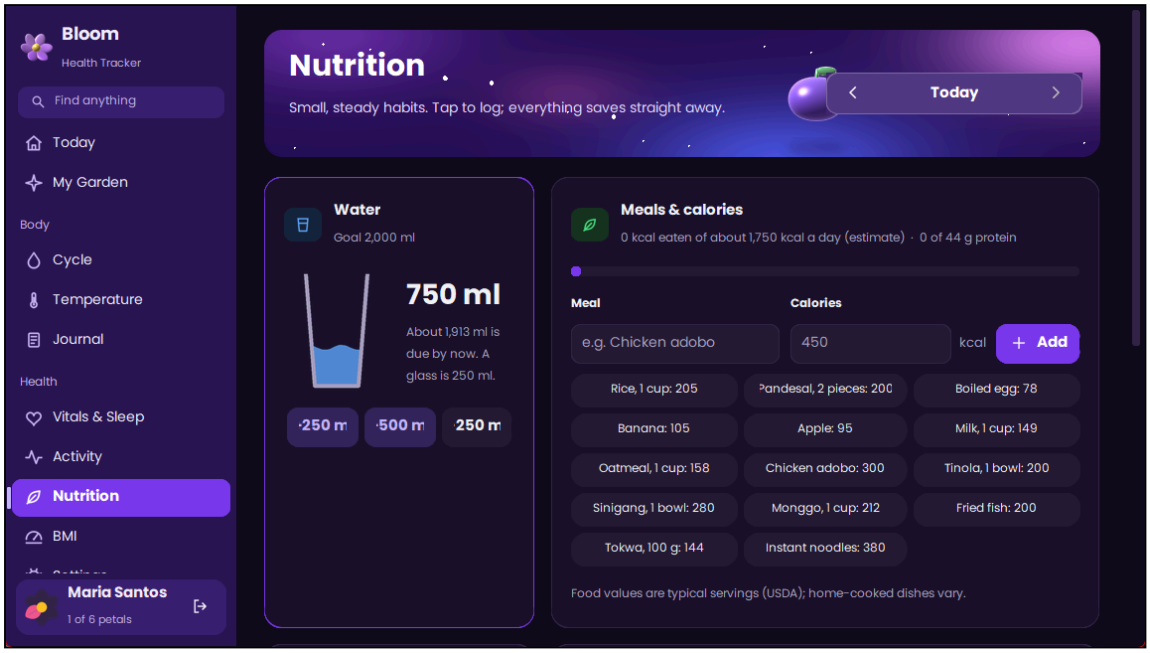
7.9 Activity

Log steps and workouts. Calories are estimated from MET values, and the week is measured against the WHO goal for your age (vigorous minutes count double), with muscle-strengthening days.



7.10 Nutrition

Add glasses of water, common foods in one tap or your own meals, caffeinated drinks, alcohol and fruit and vegetable portions. Each has a goal or limit for your age from a health guideline, and the page lists your daily iron, calcium and folate needs.



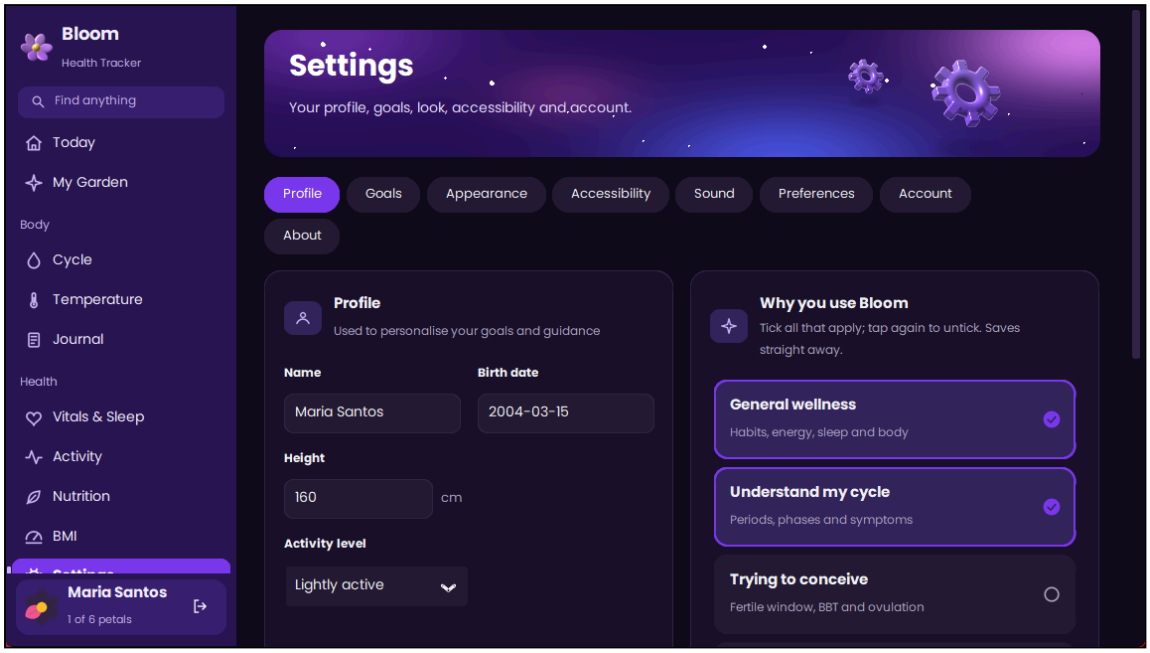
7.11 BMI

Log your weight to see your BMI on a gauge with its category, your healthy weight range and your weight history. Adults choose CDC or Asia-Pacific categories; under 20 the app uses the CDC growth charts.



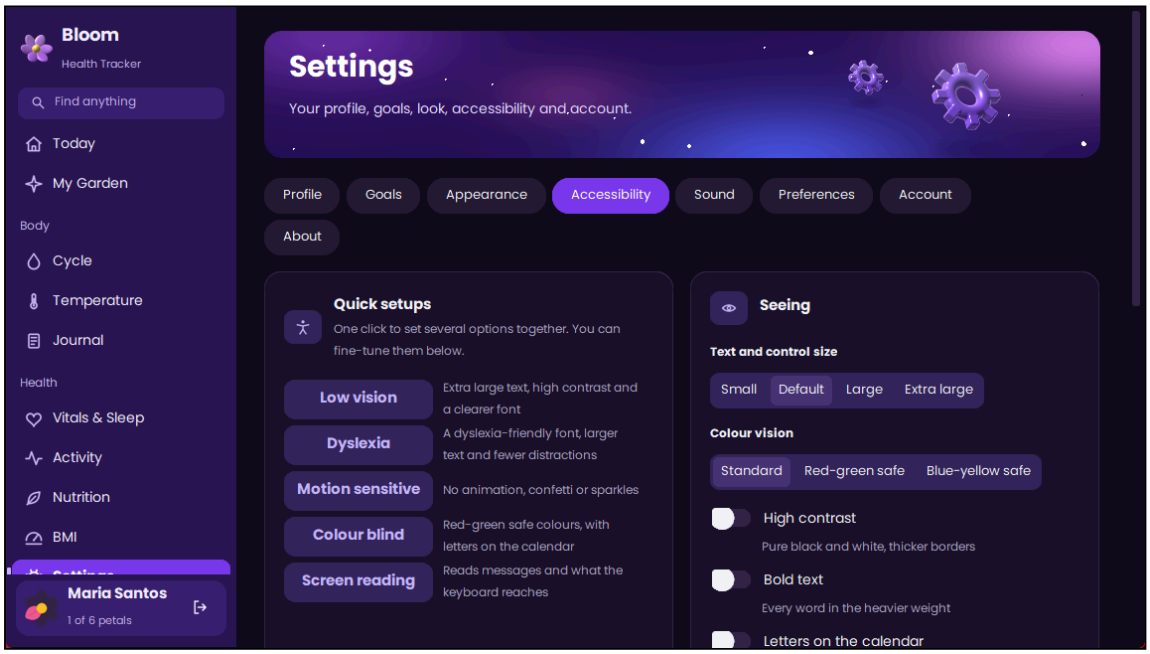
7.12 Settings

Change your profile, goals, units, colours, fonts, effects and sounds; export your data as CSV or create a printable doctor summary; change your password or recovery code; log out or delete your account; and see the sources.



7.13 Accessibility

One-click setups for low vision, dyslexia, motion sensitivity, colour blindness and screen reading, plus each option on its own: four text sizes, high contrast, colour-blind safe colours, easier-to-read fonts, reduce motion, reading aloud and more. Everything also works from the keyboard: F1 lists the shortcuts.



8. HEALTH STANDARDS USED

Topic	Value used in the program	Source
BMI formula	weight (kg) ÷ height (m)², rounded to 1 decimal	CDC
BMI categories (CDC)	<18.5 underweight; 18.5–24.9 healthy; 25–29.9 overweight; ≥30 obesity (classes 1, 2, 3 from 30, 35, 40); adults aged 20+	CDC
BMI under 20	Percentile for age: <5th underweight; 5th to <85th healthy; 85th to <95th overweight; ≥95th obesity	CDC growth charts (2000)
BMI categories (Asia-Pacific)	<18.5; 18.5–22.9 healthy; 23–24.9 overweight; 25–29.9 obesity I; ≥30 obesity II	WHO Western Pacific Region (2000)
Normal cycle length	24–38 days; under 20, 21–45 days	FIGO (2018); ACOG Committee Opinion 651
Regular cycles	shortest to longest within 7–9 days, by age; periods up to 8 days	FIGO (2018)
Luteal phase	about 14 days (ovulation = next period − 14)	NHS; Office on Women's Health
Fertile window	calendar method: shortest − 18 to longest − 11, with a margin of 1–2 days until 4 cycles are known	Office on Women's Health; Section 9
Thermal shift	3 readings above the highest of the previous 6; 3rd ≥ 0.2 °C higher, or a 4th above the coverline	Symptothermal method (Sensiplan)
LH test	ovulation usually 24–36 hours after the first positive	Office on Women's Health
Fever	38.0 °C (100.4 °F) or more	CDC
Blood pressure	Normal <120/<80; Elevated 120–129/<80; Stage 1 130–139 or 80–89; Stage 2 ≥140 or ≥90; Severe >180 or >120; Low <90 or <60	ACC/AHA (2017), CDC; American Heart Association
Resting heart rate	60–100 bpm	American Heart Association
Blood oxygen	95–100% typical; 93–94% re-measure and ask for advice; 92% or less urgent	NHS England
Sleep	9–12 hours at 6–12 years; 8–10 at 13–17; 7 or more from 18	CDC
Physical activity	Adults 150 moderate minutes a week, vigorous counts double, strength on 2 days; under 18, 60 minutes a day	WHO (2020)
Workout calories	MET × 3.5 × kg ÷ 200 per minute	Compendium of Physical Activities
Daily calories	Women: 10 × kg + 6.25 × cm − 5 × age − 161, × activity factor 1.2–1.9	Mifflin-St Jeor (1990)
Daily calories, ages 9–18	135.3 − 30.8 × age + PA × (10 × kg + 934 × m) + 25; PA 1.0–1.56	Institute of Medicine (2005)
Protein	0.8 g per kg; 0.85 at 14–18; 0.95 at 9–13	US Dietary Reference Intakes
Iron, calcium, folate	Women 19–50: 18 mg, 1,000 mg, 400 mcg; ages 14–18: 15 mg, 1,300 mg, 400 mcg; ages 9–13: 8 mg, 1,300 mg, 300 mcg	NIH Office of Dietary Supplements

Caffeine	up to 400 mg a day; 200 mg when trying to conceive; 100 mg under 18	FDA; ACOG; American Academy of Pediatrics
Alcohol	Less is better; 1 drink a day kept as the marker; not advised under 18	Dietary Guidelines for Americans 2020–2025 and 2025–2030
Fruit and vegetables	400 g (5 portions) a day	WHO
Water goal	33 ml per kg, rounded to a 250 ml glass (a design choice the user can change)	—
Password storage	PBKDF2-HMAC-SHA256, 600,000 iterations, 16-byte salt	OWASP

9. CHECKING THE CALCULATIONS ON REAL DATA

Every calculation was compared with its source, and the cycle predictions were tested on **1,665 real cycles from 159 women** (Fehring's public dataset, Marquette University). Each method predicted each woman's next cycle using only her earlier cycles. The dataset was used only for this test and is not part of the app. The full list of verdicts is in `docs/CALCULATIONS.md`.

9.1 Next period (1,334 predictions)

Method	Average error	Within 2 days	Within 3 days
Always 28 days	2.81 days	60%	73%
Same as the last cycle	2.80 days	61%	73%
Average of the last 3	2.24 days	69%	80%
Average of the last 6 (Bloom)	**2.13 days**	**70%**	**81%**
Last 6 without the highest and lowest	2.11 days	72%	81%

The best method beat Bloom's by only 0.02 days, so the simpler average stayed.

9.2 When this cycle's ovulation is known (1,238 predictions)

Method	Average error	Within 2 days
The calendar alone, on the same cycles	2.04 days	
Ovulation + 14 days	1.70 days	81%
Ovulation + the woman's own usual gap	**1.25 days**	**88%**

9.3 Fertile window: how often the real ovulation day fell inside it

Cycles known	Calendar method alone	With Bloom's margin
1	68%	90%
2	88%	94%
3	88%	93%
4	95%	95%
6 or more	97%	97%

9.4 Confidence labels

Label	Average error	Within 3 days
High	1.57 days	90%
Medium	2.21 days	80%
Low	3.41 days	61%

The labels mean what they say. The date range shown for the next period contained the real start 80% of the time. **Limits:** this dataset comes from women using natural family planning, so results may differ for other groups, and its ovulation days are themselves estimates.

10. ACCESSIBILITY

Settings has a whole Accessibility tab, with one-click setups for low vision, dyslexia, motion sensitivity, colour blindness and screen reading.

Need	What Bloom offers
Low vision	Four text sizes (also Ctrl+plus and Ctrl+minus), high contrast at 7:1 or more, bold text, larger click targets, Atkinson Hyperlegible
Colour blindness	Red-green and blue-yellow safe palettes (Okabe-Ito colours), and letters on the calendar so colour is never the only cue
Dyslexia and reading	OpenDyslexic and Lexend fonts, and messages that stay on screen longer
Motion sensitivity	Reduce motion (also taken from the Windows setting for new accounts) and hiding decorative art
No mouse	Tab with a bright focus ring, Enter or Space to use a control, arrow keys on the calendar, F6 between the sidebar and the page, Ctrl+K to find anything, F1 for the list of shortcuts
Blindness and low vision	F2 reads the page aloud, and Bloom can read messages and each control the keyboard reaches, using the voice built into the computer

Contrast is not assumed: `tests/test_look.py` calculates the WCAG ratio of every text and background pair in every theme. The first run of that test found real failures in the original colours, such as red text on a pale red background at 4.1:1 (below the 4.5:1 minimum), which were then fixed.

11. REFERENCES

1. Centers for Disease Control and Prevention. *Adult BMI Categories.* https://www.cdc.gov/bmi/adult-calculator/bmi-categories.html
2. WHO Western Pacific Region, IASO and IOTF (2000). *The Asia-Pacific Perspective: Redefining Obesity and Its Treatment.*
3. Centers for Disease Control and Prevention. *CDC Growth Charts Data Files* (BMI-for-age). https://www.cdc.gov/growthcharts/cdc-data-files.htm
4. Munro, M. G., et al. (2018). The two FIGO systems for normal and abnormal uterine bleeding symptoms. *International Journal of Gynecology & Obstetrics, 143*(3), 393–408.
5. ACOG Committee Opinion No. 651 (2015). Menstruation in Girls and Adolescents: Using the Menstrual Cycle as a Vital Sign. *Obstetrics & Gynecology, 126*(6).
6. Fehring, R. J., Schneider, M., and Raviele, K. (2006). Variability in the Phases of the Menstrual Cycle. *JOGNN, 35*(3). Data: https://epublications.marquette.edu/data_nfp/7
7. Office on Women's Health. *Your Menstrual Cycle.* https://www.womenshealth.gov/menstrual-cycle/your-menstrual-cycle
8. Office on Women's Health. *Trying to Conceive.* https://www.womenshealth.gov/pregnancy/you-get-pregnant/trying-conceive
9. NHS. *Fertility in the Menstrual Cycle.* https://www.nhs.uk/conditions/periods/fertility-in-the-menstrual-cycle/
10. Whelton, P. K., et al. (2018). 2017 ACC/AHA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults. *Hypertension, 71*(6).
11. Centers for Disease Control and Prevention. *About Sleep: How Much Sleep Do I Need?* https://www.cdc.gov/sleep/about/index.html
12. World Health Organization (2020). *WHO Guidelines on Physical Activity and Sedentary Behaviour.*
13. Ainsworth, B. E., et al. *Compendium of Physical Activities.*
14. Mifflin, M. D., St Jeor, S. T., et al. (1990). A new predictive equation for resting energy expenditure in healthy individuals. *American Journal of Clinical Nutrition, 51*(2), 241–247.
15. U.S. Food and Drug Administration. *Spilling the Beans: How Much Caffeine Is Too Much?*
16. U.S. Department of Agriculture and HHS. *Dietary Guidelines for Americans, 2020–2025* and *2025–2030.*
17. Institute of Medicine (2005). *Dietary Reference Intakes for Energy, Carbohydrate, Fiber, Fat, Fatty Acids, Cholesterol, Protein, and Amino Acids.*
18. NIH Office of Dietary Supplements. *Iron: Fact Sheet for Health Professionals.* https://ods.od.nih.gov/factsheets/Iron-HealthProfessional/
19. NHS England. *COVID Oximetry @home.* https://www.england.nhs.uk/nhs-at-home/covid-oximetry-at-home/
20. W3C. *Web Content Accessibility Guidelines (WCAG) 2.2.* https://www.w3.org/TR/WCAG22/

21. World Health Organization. *Healthy Diet* fact sheet.
22. OWASP. *Password Storage Cheat Sheet.* https://cheatsheetseries.owasp.org/cheatsheets/Password_Storage_Cheat_Sheet.html
23. Python Software Foundation. *Python 3 Documentation* (datetime, calendar, hashlib, secrets, json). https://docs.python.org/3/
24. Schimansky, T. *CustomTkinter Documentation.* https://customtkinter.tomschimansky.com